

EXPERIMENT – 8

AIM

To discover live hosts using ARP, ICMP, TCP, and UDP Nmap scans in the authorized TryHackMe lab.

THEORY

Nmap is an open-source network discovery and security auditing tool. Host discovery identifies active systems before port scanning.

Methods include:

- ARP Scan
- ICMP Echo, Timestamp, and Address Mask Scans
- TCP SYN and ACK Ping
- UDP Ping

ARP operates at the link layer, ICMP at the network layer, and TCP/UDP at the transport layer.

REQUIREMENTS

- Nmap
- Internet connection
- Authorized TryHackMe Nmap lab
- Terminal / Command Prompt

TARGET SPECIFICATION

Targets may be specified as:

TARGET

10.11.12.15-20

MACHINE_IP/30

Read targets from a file with:

`nmap -iL list_of_hosts.txt`

Use `-sL` to list targets without scanning and `-n` to disable DNS resolution.

PROCEDURE

1. ARP Host Discovery

Used on the same subnet:

`nmap -PR -sn TARGET`

`-PR` enables ARP discovery and `-sn` disables port scanning.

2. ICMP Echo Discovery

nmap -PE -sn TARGET

An ICMP Echo Reply indicates that the host is reachable. Firewalls may block this traffic.

3. ICMP Timestamp Discovery

nmap -PP -sn TARGET

Uses ICMP Timestamp Requests.

4. ICMP Address Mask Discovery

nmap -PM -sn TARGET

Uses ICMP Address Mask Requests.

5. TCP SYN Ping

nmap -PS23 TARGET

A SYN/ACK or RST response indicates that the host is reachable.

6. TCP ACK Ping

nmap -PA80 TARGET

Usually produces an RST response and requires a privileged account.

7. UDP Ping

UDP discovery may receive an ICMP Port Unreachable response from a closed port, indicating that the host is active.

OBSERVATION

Scan Type	Option Purpose	
ARP	-PR	Local subnet discovery
ICMP Echo	-PE	Echo Request discovery
ICMP Timestamp	-PP	Timestamp Request discovery
ICMP Address Mask	-PM	Address Mask Request discovery
TCP SYN Ping	-PS	TCP SYN discovery
TCP ACK Ping	-PA	TCP ACK discovery
UDP Ping	-PU	UDP discovery

IMPORTANT OBSERVATIONS

- 1. ARP works only within the local subnet.**
- 2. ICMP Echo uses Echo Request and Reply messages.**
- 3. Firewalls may block ICMP traffic.**

4. TCP SYN Ping can target a specific port.
5. TCP ACK Ping commonly receives an RST response.
6. UDP Ping may receive an ICMP Port Unreachable response.
7. Nmap can discover live hosts before further scanning.

RESULT

Live host discovery using ARP, ICMP, TCP, and UDP Nmap techniques was studied and performed successfully in the authorized lab environment.